ACTIVITY – CASE ANALYSIS: AVOIDING MISLEADING PROPORTIONS

Question 1: Truncated Y-Axis in Sales Growth

1. Why is the chart misleading?

The Y-axis starts at 90 instead of 0, exaggerating the visual difference between January (95) and June (105). The increase is only about 10.5%, not close to double.

2. Violated principle

Bar charts should use a zero baseline because bar length represents magnitude. Truncating the axis distorts comparisons.

3. Improved visualization

Use a bar chart with the Y-axis starting at 0 or use a line chart to show the monthly trend.

4. Business impact

Managers may believe sales growth is much stronger than it actually is, leading to poor budgeting, inventory, or marketing decisions.

Question 2: Pie Chart with Incorrect Proportions

1. Correct percentages

Total students = 1000

• Computer Science = 420/1000 = 42%
• Electronics = 280/1000 = 28%
• Mechanical = 200/1000 = 20%
• Civil = 100/1000 = 10%

2. Why misleading?

The displayed slices (50%, 30%, 15%, 15%) do not match the actual data and incorrectly represent department sizes.

3. Better visualization

A bar chart, because it allows more accurate comparison of values.

4. Institutional impact

incorrect enrollment comparisons may affect faculty allocation, budgeting, classrooms, and future admissions planning.

Question 3: Unequal Bar Widths in Revenue Comparison

1. Graphical error

The bars have unequal widths.

2. Distortion

Viewers judge both height and area. Wider bars make Product D appear much larger than justified by the data.

3. Correct design

Use equal-width bars, a common baseline at zero, consistent spacing, and proportional heights.

4. Consequences

Investors may overestimate Product D's performance and make incorrect investment decisions.

Question 4: 3D Column Chart Distortion

1. Why misleading?

Perspective in 3D charts changes the apparent height and size of columns, making comparisons inaccurate.

2. Actual vs perceived

Actual difference: Hydro (550) vs Wind (500) = 50 GWh = 10%. Perceived difference is about 40%, greatly exaggerating the gap.

3. Better visualization

Use a simple 2D bar chart.

4. Importance

Accurate sustainability reporting supports transparent decision-making and prevents stakeholders from drawing false conclusions.

Question 5: Bubble Chart with Incorrect Scaling

1. Why misleading?

Scaling the radius directly causes the bubble area to grow with the square of the radius, exaggerating differences.

2. Correct method

Scale the bubble area proportional to the data values. Radius should be proportional to the square root of the value.

3. Better visualization

A bar chart is the clearest option for comparing disease cases across districts.

4. Public health impact
Decision-makers may overallocate resources to some districts and underallocate them to others.